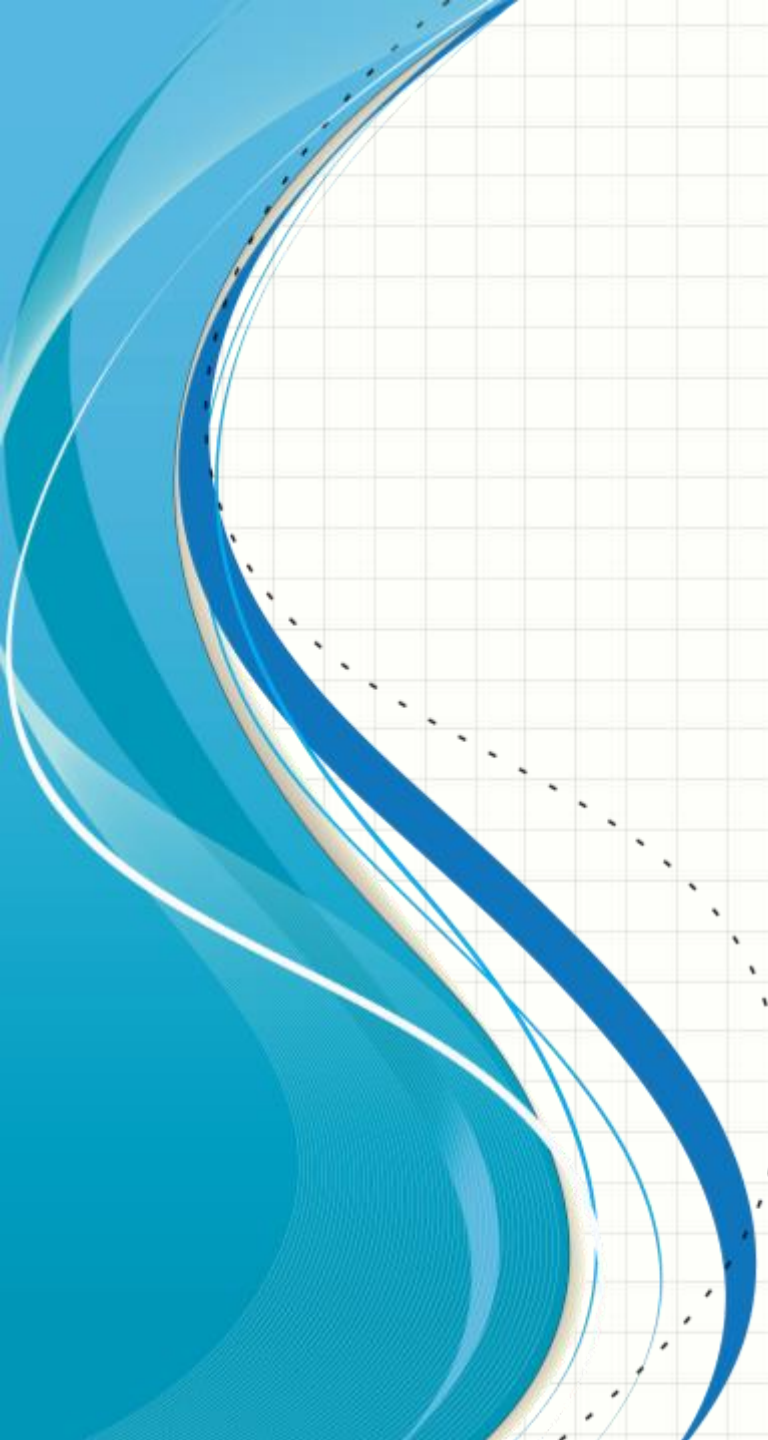




CSE-251

ELECTRONIC CIRCUITS

LECTURE-10

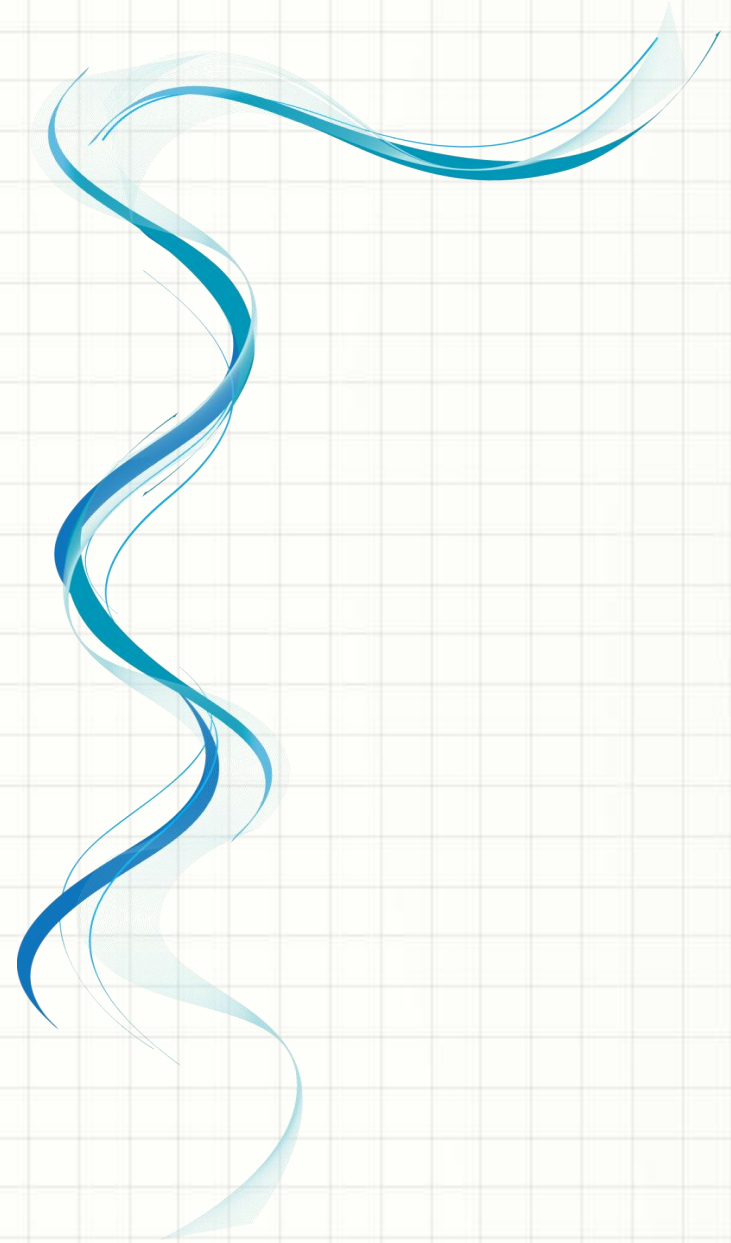
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Today's Overview:

☐ *Clampers*



Clampers



Clampers:

A clamper is a network constructed of a diode, a resistor, and a capacitor that shifts a waveform to a different dc level without changing the appearance of the applied signal.

The clamper circuit consists of a capacitor and diode connected in parallel across the load. The capacitor must be chosen such that, during the conduction of the diode, the capacitor must be sufficient to charge quickly and during the non-conducting period of the diode, the capacitor should not discharge drastically.

In general, clampers are classified into two types:

- *Positive clampers*
- *Negative clampers*

Clampers:

□ Operating Principle of Clamper Circuit:

The working of clamper circuits depends on the variation in the time constant of the capacitor. This variation is the outcome of changing the current path of the diode with the change in input signal polarity.

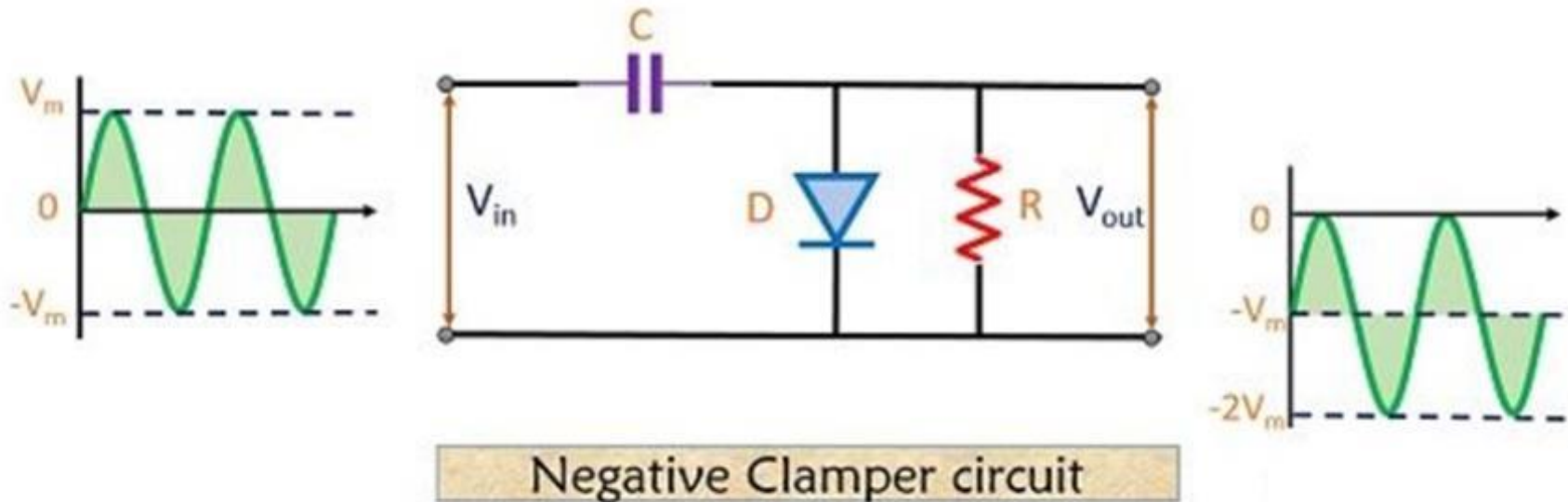
Here, the magnitude of the time constant is:

$$\tau = RC$$

This is chosen large enough in order to assure that voltage across the capacitor does not discharge consequently at the non-conducting interval of the diode.

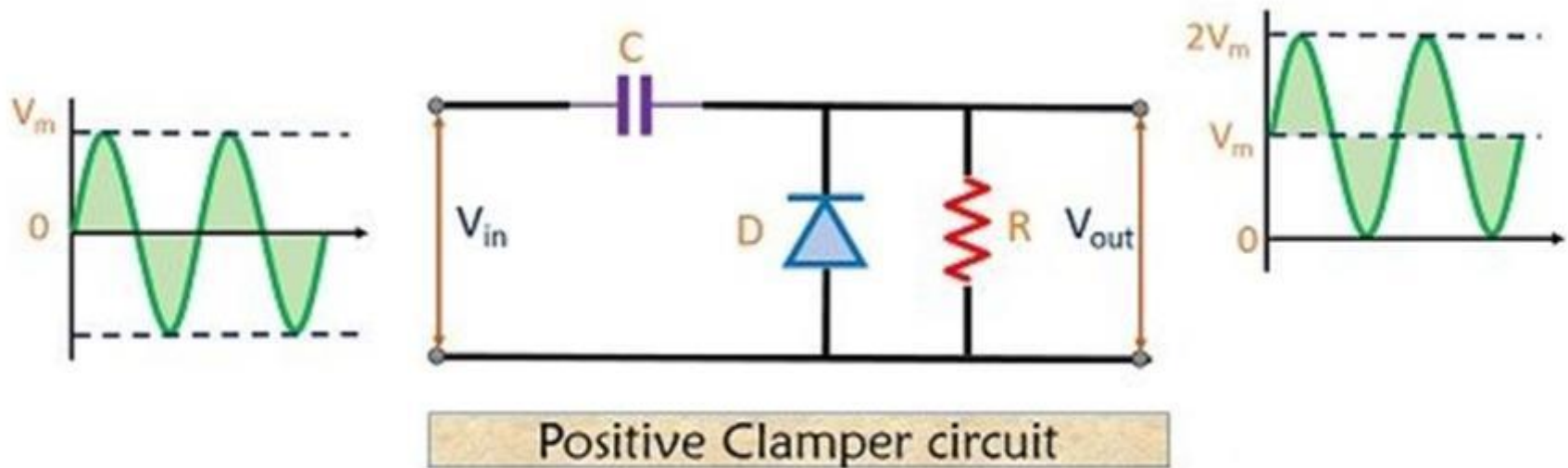
Clampers:

□ Negative Clamper:



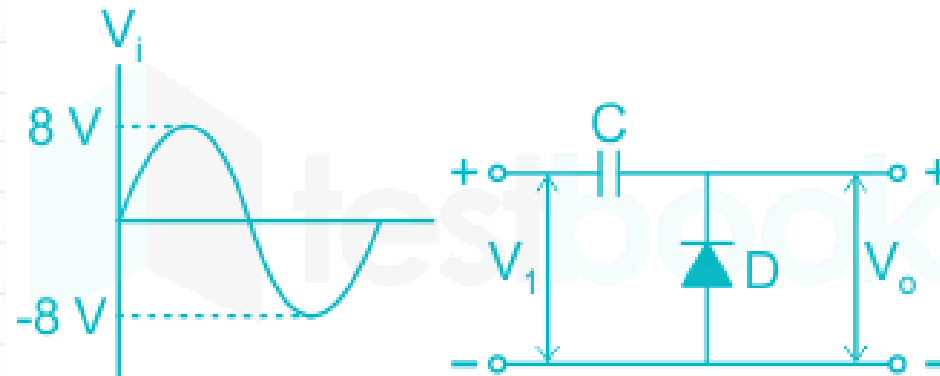
Clampers:

□ Positive Clamper:

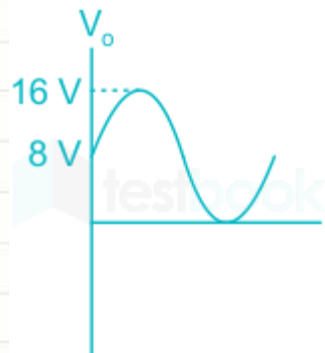


Problem:

Determine the output waveform for the given network.



Solution:





QUESTIONS?

SEE YOU IN THE NEXT CLASS

